SWIFT-U5M: 10-Step Retrospective Research Design Audit

Date: 2026-11-06 | Status: ✓ Steps 1–9 COMPLETE | Step 10: See decisions below

Step 01 — Data Inventory [VERIFIED]

All six BDHS Birth Recode (BR) files are present, unmodified, and appropriately sized.

Wave	File	Size	Date
2004	`BDBR4JFL.dta`	42.4 MB	2026-10-28
2007	`BDBR51FL.DTA`	37.2 MB	2026-10-28
2011	`BDBR61FL.DTA`	54.5 MB	2026-10-28
2014	`BDBR72FL.DTA`	56.6 MB	2026-10-28
2017-18	`BDBR7RFL.DTA`	66.9 MB	2026-10-28
2022	`BDBR81FL.DTA`	87.9 MB	2026-10-28

Source ZIPs also present in `data/zips/` (4.1–8.0 MB compressed). No files were renamed or moved.

Step 02 — Dataset Suitability [ALL PRESENT]

Every required variable was verified present in all six BR files:

Variable	Role	Present in all 6 waves
`b5`	Child alive (yes/no)	✓
`b7`	Age at death in months	✓
`bord`	Birth order	✓
`b0`	Multiple birth indicator	✓
`b4`	Child sex	✓
`b11`	Preceding birth interval	✓
`v005`	Sampling weight	✓
`v021`	Primary sampling unit (PSU)	✓
`v022`	Stratum	✓
`v012`	Respondent's current age	✓
`v025`	Residence type (urban/rural)	✓
`v024`	Region/division	✓
`v106`	Maternal education level	✓
`v190`	Wealth index	✓

v005 scaling note: Raw values > 1000 in all waves → divided by 1,000,000 ✓ (implemented in `cohort.py`)

PSU/Stratum counts per wave:

Wave	PSU unique	v022 strata	v023 strata
2004	361	177	1
2007	361	179	22
2011	600	20	20
2014	600	20	21
2017-18	672	22	22
2022	674	16	16

NOTE / IMPLEMENTATION CONTEXT: Stratum variable differs by wave: 2004 uses `v022` (177 strata); 2011–2022 use `v022`/`v023` (16–22 strata). The pipeline harmonizes via the `strata` column in the processed parquet. This is correct and documented.

Step 03 — Cohort Eligibility Audit [COMPLETE]

Total harmonized cohort: 51,489 records across 6 waves

All 51,489 records have `is_exposure_complete = 1` — the cohort was already correctly filtered to children aged 0–59 months at interview with known survival status.

Wave	N	Deaths	Prev% (unweighted)	Prev% (weighted)	Age range (months)
2004	6,908	484	7.01%	7.19%	0–59
2007	6,150	361	5.87%	5.59%	0–59
2011	8,742	410	4.69%	4.48%	0–59
2014	7,886	319	4.05%	4.10%	0–59
2017-18	8,684	356	4.10%	4.03%	0–59
2022	13,119	408	3.11%	2.99%	0–59
TOTAL	51,489	2,338	4.54%	—	—

Split-level counts:

Role	Waves	N	Deaths	Unweighted Prev	Weighted Prev
Development	2004–2014	29,686	1,574	5.30%	5.24%
Validation	2017-18	8,684	356	4.10%	4.03%
Locked Test	2022	13,119	408	3.11%	2.99%

Key observation: Secular mortality decline clearly visible: 7.01% (2004) → 2.99% (2022, weighted). This confirms temporal drift and justifies Module C recalibration.

CRITICAL DESIGN PROTOCOL: The cohort uses all births 0–59 months at interview (not completed 5-year follow-up requiring ≥60 months). This is Outcome Definition A from the design PDF — the retrospective estimand (recorded death vs. no recorded death by interview). The design PDF acknowledges this is the existing approach. No change required: the estimand is transparently documented in the manuscript as retrospective.

Step 04 — Outcome Definition Comparison [DECISION MADE]

Three candidate outcomes from the research design PDF:

Definition	Population	Y=1	Advantage	Disadvantage
A (current)	All births 0–59 mo at interview	2,338 (4.54%)	Full sample, no selection	Some children not fully followed
B (≥60 mo at interview)	Children ≥60 mo at interview	Subset only	Cleaner binary estimand	Would exclude most of cohort (all age_at_survey < 60); loses ~entire dataset
C (any death)	All births, b5==0 regardless of b7	Slightly more	Inclusive	Mixes neonatal with 5+ year deaths

Decision: Retain Definition A (current).

- Definition B is impossible here — `age_at_survey_months` ranges 0–59 for all records (the DHS birth history covers births in the 5 years before interview), so ≥60 mo would exclude the entire cohort.
- Definition A is correctly described in the manuscript as a retrospective estimand: recorded death versus no recorded death by interview date.
- The 71%/85% neonatal front-loading statistics are already documented in the manuscript.

Step 05 — Predictor Harmonization Matrix [VERIFIED]

15 candidate features from `build_features.py::CANDIDATE_FEATURES`, all present in harmonized parquet:

Feature	Type	Missing	Timing	Notes
`maternal_age_at_birth`	float64	0%	At birth	Derived: v008 − v011 − b3; unaffected by child outcome
`maternal_education`	float64	0%	At interview	Interview-time; labeled as such in Table 2
`wealth_quintile`	int64	0%	At interview	Interview-time; could change post-birth
`residence_urban`	int64	0%	At interview	Interview-time
`birth_order`	int64	0%	At birth	Structural; leakage-free

Feature	Type	Missing	Timing	Notes
parity_before	int64	0%	At birth	Structural
is_first_birth	int64	0%	At birth	Structural
preceding_birth_interval	float64	0%	At birth	First-birth → median imputed
is_twin	int64	0%	At birth	From b0
anc_visits	float64	37.6%	Antenatal	Missingness indicator present ✓
tetanus_toxoid	float64	56.0%	Antenatal	Missingness indicator present ✓
iron_supplementation	float64	60.5%	Antenatal	Missingness indicator present ✓
facility_delivery	float64	30.5%	At birth	Missingness indicator present ✓
caesarean_delivery	float64	28.3%	At birth	Missingness indicator present ✓
birth_size	float64	63.5%	At birth	Missingness indicator present ✓

Hard exclusions verified absent from feature matrix: b5, b7, v005, v021, v022, v023 ✓

Leakage controls in cohort.py:

```
LEAKAGE_COLUMNS = ['b4', 'b19', 'm4', 'm5', 'v404', 'v201', 'h1'–'h10']
```

Assert statement enforces this at cohort build time ✓

NOTE / IMPLEMENTATION CONTEXT: High missingness in iron_supplementation (60.5%) and birth_size (63.5%) is wave-dependent (earlier waves lacked these questions). Missingness indicator features (_missing columns) allow the model to use missingness patterns as a signal without discarding records.

Step 06 — Survey Weighting Audit [VERIFIED]

v005 normalization: Correctly divided by 1,000,000 in cohort.py → sample_weight column.

Scaled weight range: 0.056 – 14.19 (expected for DHS; large range reflects cluster oversampling)

Wave	N	Sum(weights)	Mean(weight)	PSU unique	Strata unique
2004	6,908	7,001.7	1.014	361	177
2007	6,150	6,057.6	0.985	361	179
2011	8,742	8,788.9	1.005	600	20
2014	7,886	8,091.8	1.026	600	20
2017-18	8,684	8,708.5	1.003	672	22
2022	13,119	13,169.9	1.004	674	16

Within-wave weight sums ≈ N (each wave's weights approximately sum to N) → confirms DHS self-weighting design per wave.

Module A weighting approach (module_a_weight.py):

1. Within-round survey weight normalization: w_survey = v005/mean(v005) per round ✓
2. Exponential time-decay: w_time = exp(−λ × (T_target − T_round)) with λ=0.05 ✓
3. Density ratio shift correction (clipped domain classifier) ✓
4. Combined: W_i = w_survey × w_time × w_shift, normalized to mean=1 ✓

Bootstrap uncertainty (bootstrap.py): PSU-resampling within strata — correctly implements DHS design-aware cluster bootstrap ✓

Estimand: Retrospective survey-weighted risk classification within each wave's nationally representative sample. The six waves are pooled for training (not merged into a single population-representative sample). This is correctly described in the manuscript.

Step 07 — Temporal Split [FROZEN & APPROVED]

Role	Waves	N	Deaths
Development	2004, 2007, 2011, 2014	29,686	1,574
Temporal Validation	2017-18	8,684	356
Locked Test	2022	13,119	408

ROUND_ORDER = ['2004', '2007', '2011', '2014', '2017-18', '2022'] in splits.py ✓

Rolling-origin forward-chaining cuts (Table 3 in manuscript): 5 cuts (2004→2007, 2004-07→2011, etc.) implemented in `get_forward_chaining_cuts()` ✓

The 2022 test set outcomes have NOT been used for tuning. All module calibration uses 2017-18 validation data.

Step 08 — Implementation Audit (Read-Only) [NO DISCREPANCIES]

Audited in order specified by the research design PDF:

1. `src/data/cohort.py`

- Eligibility: births 0–59 months before interview ✓
- Outcome: `(b5 == "no") & (b7 < 60)` ✓
- Leakage columns asserted absent ✓
- v005 /1e6 scaling ✓
- Zero/negative weight records excluded ✓

2. `src/data/harmonize.py` / `dataset.py`

- 322 lines; loads and harmonizes all 6 waves
- Applies `filter_exposure_complete_cohort()` per wave ✓
- Produces `u5m_harmonized_cohort.parquet` ✓

3. `src/validation/splits.py`

- `ROUND_ORDER` matches approved temporal split ✓
- `get_forward_chaining_cuts()` produces 5 rolling-origin cuts ✓
- `encode_features()`: division one-hot, train-only imputation by median ✓
- `compute_normalized_weights()`: normalizes to mean=1 ✓

4. `src/features/select.py`

- Dual-method selection: Weighted LASSO + XGBoost permutation importance ✓
- SHAP excluded from feature selection (quarantine rule; test `test_zero_shap_leakage_in_selection` passes) ✓
- High-missingness (>30%) and high-collinearity (|r|>0.8) filter applied within training data only ✓

5. `src/models/module_a_weight.py`

- Survey + time decay + density ratio weights ✓
- Domain classifier fit on train vs. target round features ✓
- Clipped: [0.2, 5.0] ✓

6. `src/models/module_b_shrink.py`

- Empirical Bayes shrinkage: `λ_d = D_d / (D_d + κ)`, κ=40 ✓
- National head fit on out-of-fold predictions ✓
- Division fallback if deaths < 5 or n < 30 ✓

7. `src/models/module_c_recalib.py`

- Platt recalibration on 2017-18 (most recent pre-deployment round) ✓
- SDI computed as mean absolute difference in feature-level mean |SHAP| ✓
- Test outcomes (2022) NOT used for calibration ✓

8. `src/evaluation/metrics.py`

- Weighted ROC-AUC, PR-AUC, Brier, ECE ✓
- Calibration slope/intercept via logistic recalibration ✓
- Screening metrics at 5%, 10%, 20% capacity ✓

9. `src/evaluation/bootstrap.py`

- PSU-stratified cluster bootstrap (DHS design-aware) ✓
- Precomputed cluster maps for speed ✓
- Current n_replicates=200 (see Step 9 below for 1000-replicate upgrade)

10. Config and runner

- `RANDOM_SEED = 42` ✓
- Target round specified at fit time, not hardcoded ✓

Summary discrepancy log:

Component	Issue	Severity	Action
`bootstrap.py`	200 replicates (design recommends assessing 1000)	Minor	Upgrade to 1000 in Step 10
None others	—	—	—

Step 09 — Statistical Analysis Plan [FROZEN]

Primary Metrics (weighted, 2022 test set)

- Discrimination: ROC-AUC (W), PR-AUC (W)
- Classification: Accuracy, Precision, Recall/Sensitivity, Specificity, F1, Balanced Accuracy, Confusion matrix at Youden-optimal threshold (τ = 0.0004)
- Probability quality: Brier Score (W), Brier Skill Score (BSS), Calibration intercept (α), Calibration slope (β), ECE, Reliability diagram
- Population vulnerability stratification: Survey-weighted event capture, Sensitivity, PPV, NNS, Deaths/1k screened at 5%, 10%, 20% capacity (rank-cutoff and weighted-capacity regimes)
- Uncertainty: PSU-stratified cluster bootstrap, 1000 replicates, 95% CIs (2.5th–97.5th percentile)

Benchmark Numbers (Frozen — not to be "improved")

Metric	Value
ROC-AUC (SWIFT-U5)	0.637 [0.600, 0.676]
ROC-AUC (XGBoost)	0.646
ROC-AUC (Random Forest)	0.645
PR-AUC	0.0887
Null Brier	0.0290
SWIFT-U5 Brier	0.0297
BSS	−0.024
Cal intercept (α)	+0.032
Cal slope (β)	0.425
Youden τ	0.0004
Sensitivity (Youden)	64.2%
Specificity (Youden)	56.8%
NNS (10% rank)	12.1
NNS (10% weighted cap)	12.8

Subgroup Analyses

- Residence: rural/urban
- Wealth: quintiles Q1–Q5
- Maternal education: 4 levels (none, primary, secondary, higher)
- Division: all 7+1 administrative divisions

Ablation Design

Model	Modules
Full SWIFT-U5	A + B + C
−A	B + C only
−B	A + C only
−C	A + B only
Base XGBoost	No modules

Sensitivity Analyses

1. Rolling-origin forward-chaining temporal cuts (Cut 1–5)
2. Age-at-interview bands (0–11, 12–35, 36–59 months)
3. SHAP Drift Index by wave pair
4. Geographic subgroups by division

Bootstrap Upgrade (Step 10 action)

- Current: 200 replicates
- Target: 1,000 replicates
- Feasibility: [CONFIRMED] — 674 PSUs, 16 strata, precomputed maps → <60 seconds
- Impact on existing CIs: Minor widening expected (more stable tail estimates)

Step 10 — Planned Pipeline Revisions

> CRITICAL DESIGN PROTOCOL: Only one code change is required: upgrade bootstrap from 200 → 1000 replicates. All other components match the approved design.

Actions

1. Update `bootstrap.py` default: Change `n_replicates: int = 200` → `n_replicates: int = 1000`
2. Re-run bootstrap CIs on the 2022 test set to get final 1000-replicate confidence intervals
3. Update manuscript tables with new CIs if materially different
4. Run all 21 tests to confirm passing
5. Rebuild Overleaf zip and push to git

What is NOT changing

- Cohort construction (already correct)
- Outcome definition (Definition A, retrospective — correct)
- Feature set (15 candidates, dual-method selection)
- Model modules A, B, C (all correct)
- Temporal split (frozen: dev 2004–14, val 2017-18, test 2022)
- All benchmark numbers (treating as manuscript benchmarks per design PDF §7)

Tests Status

```
Ran 21 tests in 0.515s — OK ■
```

Summary Verdict

Step	Finding	Status
01 Data inventory	6 BR files, 37–88 MB, all present	[PASS]
02 BR suitability	All 14 required vars in all 6 waves	[PASS]
03 Cohort eligibility	51,489 records; all exposure-complete	[PASS]
04 Outcome definition	Definition A (retrospective); B impossible	[RETAINED]
05 Predictor harmonization	15 features; missingness indicators present	[PASS]
06 Survey weighting	v005 /1e6; PSU-stratified bootstrap; dual normalization	[PASS]
07 Temporal split	Dev 2004-14, Val 2017-18, Test 2022 frozen	[FROZEN]
08 Code audit	No critical discrepancies; 1 minor (bootstrap n)	[PASS]
09 SAP frozen	Metrics, CIs, subgroups, ablations, sensitivity	[FROZEN]
10 Revisions	Bootstrap 200→1000 only	[PENDING]